

George Shenoda

Engineering Student | IoT & Embedded Systems Enthusiast

CONTACT

- **Phone:** +20 1278682171
- **Email:** georgeshenoda@protonmail.com
- **Address:** Ain Shams, Cairo, Egypt
- **LinkedIn:** [linkedin.com/in/george-shenoda-iot](https://www.linkedin.com/in/george-shenoda-iot)
- **GitHub:** <https://george-iot.github.io/>
- **Certifications Portfolio:**
<https://drive.proton.me/urls/7C3Q1MYZA8#Zkn1spNBJDts>

SUMMARY

A proactive sophomore Mechatronics Engineering student seeking an IoT or Embedded Systems Internship. Fresh from a 3-month internship at the Samsung Innovation Campus, I have hands-on experience in system integration, successfully developing a smart office solution that combines facial recognition, NFC, and multiple environmental sensors. Proficient in Python, C++, and various microcontrollers, I am eager to apply my skills to solve complex engineering problems.

EXPERIENCE

Samsung Innovation Campus | Cairo, Egypt *IoT Intern* | Aug 2025 - Oct 2025

- Developed and deployed 2 end-to-end IoT prototypes, including a multi-factor smart office access system, during an intensive 3-month program.
- Programmed custom C++ firmware for ESP32 and Arduino microcontrollers to manage real-time data from 5+ sensors and control multiple actuators.

- Collaborated within a team of 4 interns to integrate hardware with cloud platforms, successfully presenting a final project that solved a real-world automation challenge.

EDUCATION

Ain Shams University | Cairo, Egypt *Bachelor of Engineering, **Mechatronics***

- Expected Graduation: 2028

Saint Mary Language School | Cairo, Egypt *High School Diploma*

- Graduated: 2024

PROJECTS

- **IoT Smart Office System**
 - Architected and developed a comprehensive smart office system using a Raspberry Pi as the central controller and an Arduino for sensor/actuator management.
 - Engineered a dual-factor authentication access control system combining NFC card scanning with facial recognition to verify identity and grant entry.
 - Implemented a real-time fire safety module that detects smoke, automatically activating a water pump and a buzzer, while instantly notifying the manager via WhatsApp and email using Node-RED.
 - Designed a centralized Node-RED dashboard displaying live environmental data from DHT (temperature, humidity) and gas sensors.
 - Integrated a meeting room booking system and a role-based permission model, where an "Admin NFC" tag unlocks privileged dashboard features.
- **KNX-Based Home Lighting Control**
 - Designed and simulated a smart lighting system using the KNX protocol.
 - Programmed virtual devices within ETS6 to control lighting functions, including dimming and switching via virtual push buttons.

TECHNICAL SKILLS

- **Programming Languages:** Python, **C/C++**, **Embedded C**, C#, Rust, Dart, Kotlin, Swift
- **Hardware & Microcontrollers:** Raspberry Pi, Arduino, ESP32
- **Communication Protocols:** MQTT, HTTP, CoAP, WebSocket, KNX, NFC
- **Tools, Platforms & Cloud:** Git, Node-RED, ETS6, MATLAB, Android Studio, Flutter, Blynk

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **IoT Course Completion** | Samsung Innovation Campus (Oct 2025)
- **Introduction to IoT** | Mahara Tech (Aug 2025)
- **KNX Basic Course** | Aviation Student Activity ASU (July 2025)
- **Problem Solving** | FEHU - CPC Community (Jun 2025)
- **Robotics and Automation** | IEEE Helwan Branch (Feb 2025)

LANGUAGES

- **Arabic:** Native
- **English:** B2
- **Italian:** A2